Unit 52 Green chemistry

Pre-study
A pre-study worksheet on this unit is available for download from *Jing Kung Chemistry Website*.



Fig. 52.1 Chemistry has improved virtually every aspect of our lives

52.1 Sustainable development and green chemistry

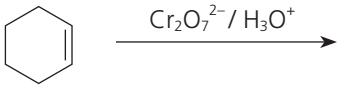
You probably notice that every aspect of our lives is dependent on chemistry. Pharmaceutical breakthroughs have greatly extended the average human lifespan. Improved agricultural products like herbicides and pesticides allow farmers to produce more food. Composites provide improved features in consumer products.

Video
Waste reduction and green chemistry: Sai Tso Wan Recreation Ground

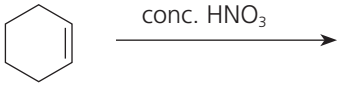
Worksheet
A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.

Discussion

Hexanedioic acid is an industrially important compound. It is the monomer used in the manufacture of nylon-6,6. The acid is commonly synthesized from cyclohexene using the following two methods:



cyclohexene → $HOOC-(CH_2)_4-COOH$ (hexanedioic acid)



cyclohexene → $HOOC-(CH_2)_4-COOH$ (hexanedioic acid)

Discuss why both methods are considered as environmentally unfriendly.

ASL 2003 Q12(a)(ii)

Method 1:
The waste contains dichromate ions which are toxic.
Removal of chromium(III) ions from the product is costly.
Method 2:
Concentrated nitric acid is a strong acid. Discharge of the waste into waterways leads to environmental pollution.

DSE 2015 Paper 2 Q2(a)(iii)

In the above discussion, you may associate chemistry with creating environmental pollution problems.

Recently the issue of the 'environmental impact' of chemicals has come into public awareness. Everyone is increasingly aware of the need to act more responsibly to protect our world for future generations. This is crystallized in the drive for **sustainable development**.

According to the United Nations,

> Sustainable development can be defined as 'development meeting the needs of the present without compromising the ability of future generations to meet their needs'.

The three components of sustainable development are economic growth, environmental protection and social equity. It is suggested that we can achieve all three by gradually changing the ways in which we develop and use technologies.



Fig. 52.2 The three components of sustainable development

Historically, the general public was not willing to place a high priority on protecting the environment when there was concern about achieving a certain level of welfare or economic goals. Now for many nations, the ability of the economy to meet basic needs allows them to focus more on environmental issues.

sustainable development 可持續發展

As governments and businesses focus on the transition to sustainability, they increasingly recognize the need to balance the often conflicting challenges of economic growth, environmental protection and social progress.

N1 Refer to page T58.

Green chemistry is chemistry for the environment. It was first N1 defined by the United States in the early 1990s as 'the utilization of a set of principles that reduce or eliminate the use or generation of hazardous substances in the design, manufacture and use of chemical products'. Green chemistry has gradually become recognized as a method for achieving sustainability.

N2 Refer to page T58.

N3 Refer to page T59.

N4 Refer to page T59.

Fig. 52.3 shows the 12 principles of green chemistry.

12 principles of green chemistry

1 Prevention — It is better to prevent waste than to treat or clean up after waste has been formed.

2 Atom economy* — When creating products it is important to maximize the incorporation of all the materials used into the final product. N2

3 Less hazardous synthesis — Processes should be designed to use and generate substances that have little to no toxicity.

4 Designing safer chemicals — Chemical products should be designed to function well while reducing toxicity.

5 Safer auxiliary substances — Whenever possible, avoid the use of auxiliary substances such as solvents and separating agents. When these substances are needed, they should be non-toxic.

6 Energy efficiency — Energy efficiency should be considered when designing and producing a product. N3

7 Use of renewable resources — When possible, the feedstocks should come from a renewable resource. N4

8 Reducing derivatives — Unnecessary derivatization (e.g. use of blocking groups) should be avoided if possible.

9 Catalysis — Catalysts speed up chemical processes and reduce the energy requirement of processes.

10 Design for degradation — Design chemicals and products that can degrade after use.

11 Use real-time analysis for pollution prevention — Methods need to be further developed to allow for real-time, in-process monitoring and control prior to the formation of hazardous substances.

12 Accident prevention — Substances and their states used in chemical processes should be chosen to minimize the potential for chemical accidents, including fires, explosions and releases.

AL 2008 Paper 1 Q10

Fig. 52.3 The 12 principles of green chemistry

green chemistry 綠色化學

*

Atom economy is a method of expressing how efficiently a particular reaction makes use of the reactant atoms.

$$\text{Atom economy} = \frac{\text{relative molecular mass or formula mass of desired product}}{\text{total relative molecular mass or formula mass of reactants}} \times 100\%$$

We have discussed this in Topic 14 Industrial Chemistry.

We may regard green chemistry as a 'reduction' process which aims at reducing materials, energy, risk and hazard, waste, cost and environmental impacts, etc.

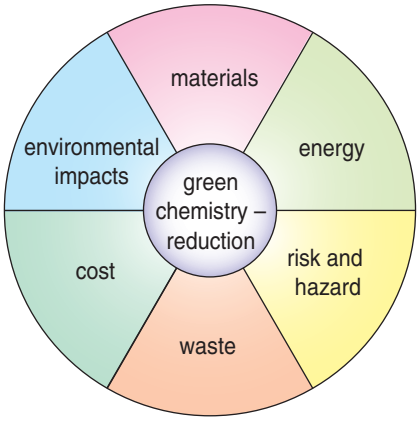


Fig. 52.4 Green chemistry can be regarded as a 'reduction' process

In the following sections, we will look at some green chemistry practices in the production of synthetic materials, including N5 N6

- the use of reactants with little or no toxicity;
- the use of catalysts; and
- the use of safer solvents and auxiliaries.

52.2 Using reactants with little or no toxicity

During the production of synthetic materials, green chemistry aims at using reactants with little or no toxicity.

Originally the company Monsanto used hydrogen cyanide in the process to produce the *weedkiller* marketed as 'Roundup'. Hydrogen cyanide is extremely toxic. The company has now developed a new route for making the weedkiller. The new method has a different starting material and runs under milder conditions.

N5 Haber process — an example of green chemistry

The Haber process is an example of green chemistry because

- the reaction has 100% atom economy;
- the reactants (N_2 and H_2) are non-toxic and pose no harm to the environment;
- the reaction takes place in gas phase (no solvent is used);
- the product is separated by cooling (no separating agent is used);
- the raw material (N_2) is available in large amount in the atmosphere (depletion is not a problem);
- the reaction does not require the use of any derivatives;
- a catalyst is used in the process.

N6 Refer to page T60.



Fig. 52.5 This weedkiller is now made by a method that does not involve the use of toxic hydrogen cyanide

weedkiller 除草劑

Similarly, a new process for making polycarbonate plastics has replaced the gas *phosgene* with safer starting materials: methanol and carbon monoxide. Carbon monoxide is toxic but it is not as dangerous as phosgene, which is so nasty that it has been used as a poison gas in warfare.

52.3 Catalysts

A catalyst alters the rate of a reaction without itself undergoing any permanent chemical changes. Catalysts are important in the development of greener chemical synthesis.

- Catalysts affect the conditions that are needed for chemical processes, often reducing the time for production and saving energy.
- Many catalysts are also highly selective, converting a high percentage of the consumed reactant to the desired product. Few or no by-products are formed, thus reducing the waste.
- Biocatalysts, enzymes produced by microorganisms, are able to catalyze chemical processes efficiently under mild conditions.

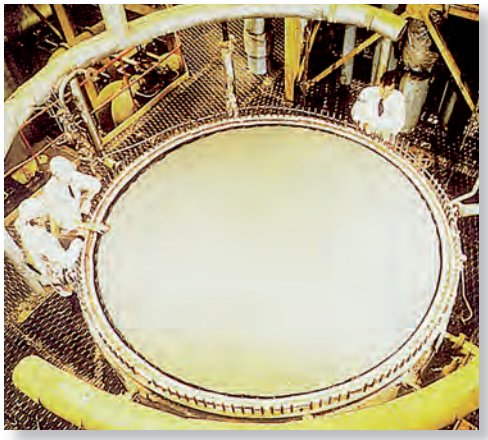


Fig. 52.6 Fitting a platinum-rhodium catalyst gauze to a reactor where ammonia will be oxidized to make nitric acid

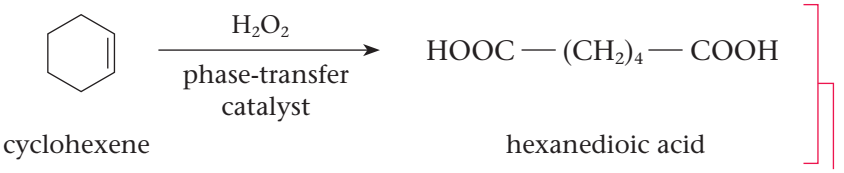


Fig. 52.7 Catalysts are made in a variety of shapes and sizes

In the discussion under 52.1, you have looked at two synthesis methods for hexanedioic acid, both of which are considered as environmentally unfriendly. Recent research shows that in the presence of a **phase-transfer catalyst**, hydrogen peroxide can be used as an oxidant in the synthesis of the acid. Hydrogen peroxide is an excellent environmentally benign oxidant, producing only water as a by-product.

DSE 2012 Paper 2 Q2(a)(ii)(2)

phosgene 光氣　　phase-transfer catalyst 相轉移催化劑



DSE 2012 Paper 2 Q2(a)(ii)(1)
ASL 2003 Q12(a)(iii)

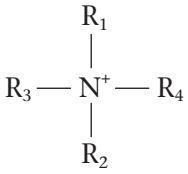
So, what is a phase-transfer catalyst? How does it work in the above synthesis?

52.4 Phase-transfer catalysis

A phase is defined as a uniform portion of matter separated from other portions of matter by a boundary surface*. A phase-transfer catalyst is a substance that will increase the rate of reaction between reactants present in separate phases. It speeds up a reaction by ensuring a ready supply of a necessary reagent to the phase in which the reaction is occurring. N7

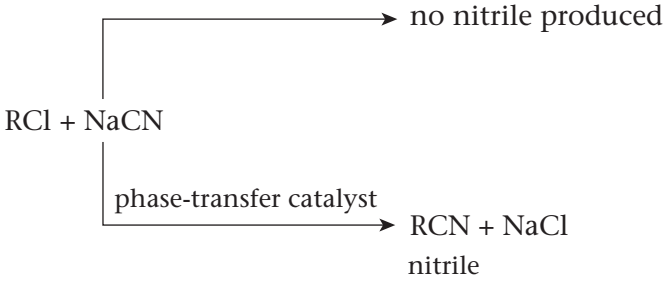
* For example, a mixture of gases is a single phase because there is no boundary surface between the different gases. However, a mixture of oil and water has a boundary surface and there are two phases.

Quaternary ammonium salts are frequently used phase-transfer catalysts. A quaternary ammonium salt is composed of a quaternary ammonium cation and an anion. A quaternary ammonium cation has the following structure:



where the $R_{1,2,3,4}$ represent alkyl groups

Consider the reaction between a chloroalkane and an aqueous solution of sodium cyanide to form a nitrile. Cyanide ions stay in the aqueous phase and the molecules of chloroalkane stay in the organic phase, so the reactants can only meet at the phase boundary. The reaction is extremely slow.



N7 Possible green advantages in using phase-transfer catalysis

There are many possible green advantages in using phase-transfer catalysis.

- Higher productivity – These reactions are often very rapid.
- Higher selectivity – These reactions can often be run at lower temperatures which may reduce by-product formation.
- Ease of product separation – The organic layer is substantially free from water-soluble contaminants. It can be decanted off and product separation is often simple.
- Use of less hazardous solvents – Simple benign solvents can often be used since phase-transfer catalysis avoids the need to find a solvent that will dissolve all reactants.

Addition of a small amount of an appropriate quaternary ammonium halide leads to a quick reaction. The quaternary ammonium halide is an ionic species having four alkyl groups. It is soluble in both the organic and aqueous phases.

Fig. 52.8 shows what happens during the catalytic process. We represent the phase-transfer catalyst (quaternary ammonium halide) by Q^+X^-. The catalyst causes the transfer of the cyanide ions as ion pairs (Q^+CN^-) into the organic phase. In the organic phase, the CN^- ions react with the chloroalkane. The Q^+ cations transport the chloride ions back to the aqueous phase. This process continues until all the reactants have reacted.

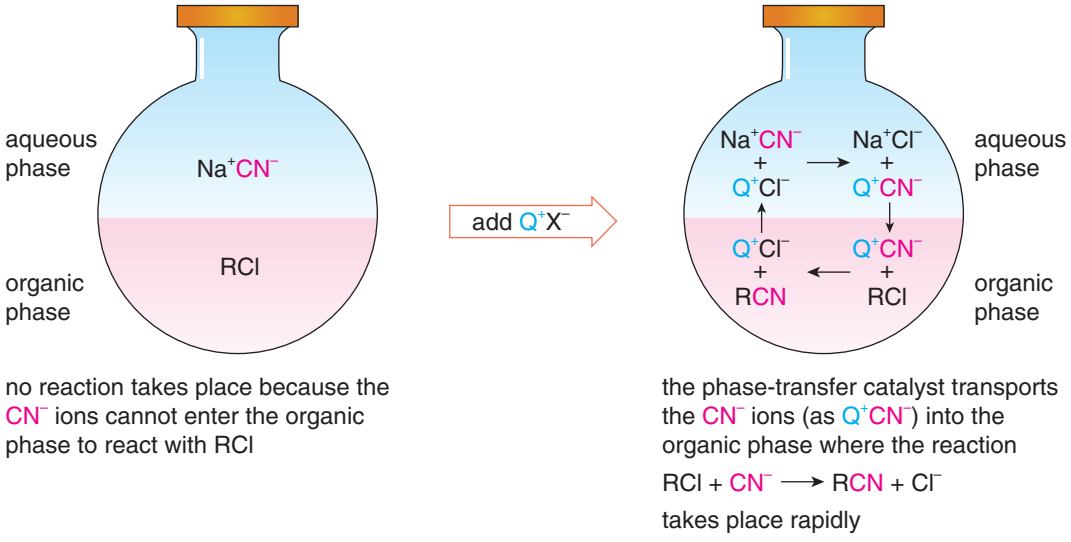
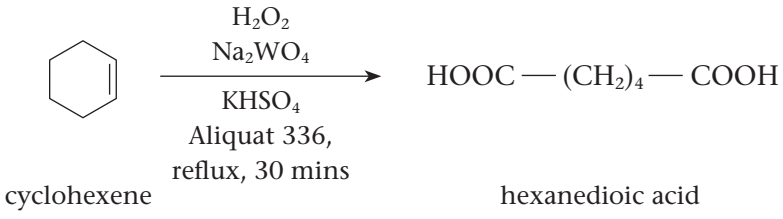


Fig. 52.8 Using a phase-transfer catalyst for the reaction between sodium cyanide and a chloroalkane

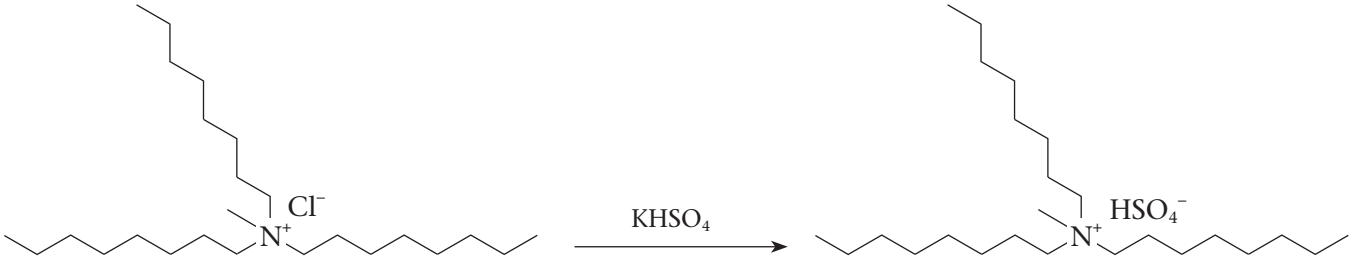
Oxidation using hydrogen peroxide

We can obtain hexanedioic acid by the oxidation of cyclohexene with a 30% aqueous hydrogen peroxide solution in the presence of small amounts of sodium tungstate (Na_2WO_4), Aliquat 336 and potassium hydrogensulphate.



The reactants in the reaction mixture are separated in two phases. Cyclohexene is in the organic phase while the oxidants (hydrogen peroxide and sodium tungstate) are in the aqueous phase. They have difficulties in coming into each other and therefore the rate of the reaction is extremely low.

Reaction of Aliquat 336 (methyltrioctylammonium chloride) with potassium hydrogensulphate yields the phase-transfer catalyst methyltrioctylammonium hydrogensulphate.



Aliquat 336
(methyltrioctylammonium chloride)

methyltrioctylammonium
hydrogensulphate

The quaternary ammonium cations of the phase-transfer catalyst form ion pairs with the HWO_4^- ions in the aqueous phase and transfer them into the organic layer to oxidize cyclohexene*. Without the phase-transfer catalyst, the reaction rate is reduced by about 100-fold.

In the following activity, you are going to prepare hexanedioic acid in the presence of a phase-transfer catalyst.

* The HWO_4^- ions come from sodium tungstate. The oxidation involves a number of steps. Sodium tungstate acts as both an oxidizing agent and a catalyst. Only the oxidizing agent hydrogen peroxide is consumed in the process.



Activity 52.1

Preparing hexanedioic acid by catalytic oxidation of cyclohexene in the presence of a phase-transfer catalyst.

Worksheet Assessment Form
A worksheet and an assessment form on this activity are available for download from *Jing Kung Chemistry Website*.



Internet Search & Presentation

The use of Ziegler-Natta catalysts in the production of polythene

The first polythene manufactured was low density polythene. Its discovery dates back to 1933. Although it was used as an insulator for cables for ground and airborne radar equipment during World War II, consumer products made of polythene only became available in 1948. Other types of polythene appeared in the 1960s, with the use of new Ziegler-Natta catalysts.

Task

Your teacher will divide the class into groups. Each group should give a 4-minute presentation on the use of Ziegler-Natta catalysts in the production of polythene.

Support your presentation with various aids.

Hints for the search Refer to page T62.

1 What are the conditions employed for the manufacture of polythene using Ziegler-Natta catalysts?

2 What type of polythene is obtained with the use of Ziegler-Natta catalysts?

3 Compare the synthesis method for polythene when it was first made* with that making use of Ziegler-Natta catalysts. Discuss which process is 'greener'.

Reference websites

1 You may start your search from the following websites:

- The story of polythene
  http://www.personal.rdg.ac.uk/~spsolley/pe.html
- A summary on production methods of polythene
  http://www.greener-industry.org.uk/pages/poly(ethene)/4_polyethene_PM_summary.htm

2 Extend your search using search engines and appropriate keyword.

- http://www.zoo.com/Zoo-Site/
- http://www.google.com
- http://www.yahoo.com
- http://www.searchalta.com/

* We have discussed the discovery of polythene in Topic 7 Fossil Fuels and Carbon Compounds. You may refer to that.

52.5 Safer solvents and auxiliaries

According to the principles of green chemistry, we should avoid the use of auxiliary substances (such as solvents and separating agents) whenever possible. When these substances are needed, they should be non-toxic.

A more environmentally sound blowing agent — carbon dioxide

Manufacturing expanded polystyrene sheet requires the use of a blowing agent. Traditionally, these blowing agents have been chlorofluorocarbons (CFCs), hydrochlorofluorocarbons (HCFCs) and aliphatic hydrocarbons. CFCs and HCFCs can contribute to both global warming and ozone depletion, while the use of aliphatic hydrocarbons can lead to the formation of photochemical smog and reduced worker safety due to their flammable nature.

Dow Chemical has developed a process to make expanded polystyrene sheet using carbon dioxide as a blowing agent*. Carbon dioxide is non-flammable, does not contribute to the formation of smog and is not ozone-depleting. The carbon dioxide used is the by-product of existing commercial and industrial sources. Hence this does not contribute to a net increase of carbon dioxide in the atmosphere. Carbon dioxide is a much safer and more environmentally sound blowing agent.

* Dow Chemical won an award for green chemistry for developing this alternative method for making expanded polystyrene.

Green solvents

Chemical processes are often carried out in media, usually organic solvents or water. As many organic reactants are insoluble in water, organic solvents are required. Volatile organic compounds (VOCs) are frequently used because of their ease of removal or evaporation.

The main environmental issue concerned with VOCs is their ability to form ozone and photochemical smog*. Using solvents such as supercritical carbon dioxide or water can make chemical processes greener.

* We have discussed how VOCs cause photochemical smog in Topic 14 Industrial Chemistry.

Reactions without solvents

Although there is a great advancement with the development of green solvents, the best alternative would be to carry out chemical processes without solvents.

In solventless processes, solid reactants can be brought together to react, either by changing their states or grinding the solids together. The advantages of carrying out chemical processes without solvents include:

- faster reaction;
- lower energy usage;
- direct formation of high purity products; and
- simplicity and low cost in equipment design.

52.6 Supercritical carbon dioxide

A substance can exist in the solid, liquid or gas phase. Both temperature and pressure can affect in which phase a particular substance exists.

A **phase diagram** summarizes the conditions under which a substance exists as a solid, liquid or gas for a closed system. Fig. 52.9 shows the phase diagram of carbon dioxide. The graph is divided into three regions, each of which represents a pure phase. The line separating any two regions indicates the conditions under which these two phases can coexist in equilibrium.

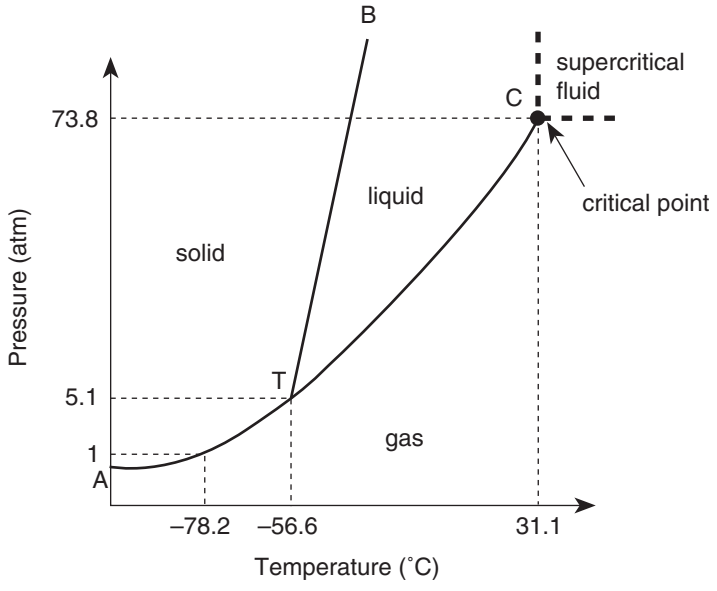


Fig. 52.9 Phase diagram of carbon dioxide

phase diagram 相圖

Line TB shows the conditions of pressure and temperature under which the solid and liquid phases are in equilibrium. Line TC shows the conditions of pressure and temperature under which the liquid and gas phases are in equilibrium. Line TA shows the conditions of pressure and temperature under which the solid and gas phases are in equilibrium.

Reference Website
Refer to the following website for video clips on supercritical carbon dioxide:
http://wn.com/supercritical_carbon_dioxide?upload_time=all_time&orderby=relevance
(accessed May 2015)

The three lines meet at a point called the **triple point** (i.e. point T) at which the three phases coexist at equilibrium.

Line TC terminates at a point called the **critical point** (i.e. point C). Beyond this point, the liquid and gas phases become indistinguishable.

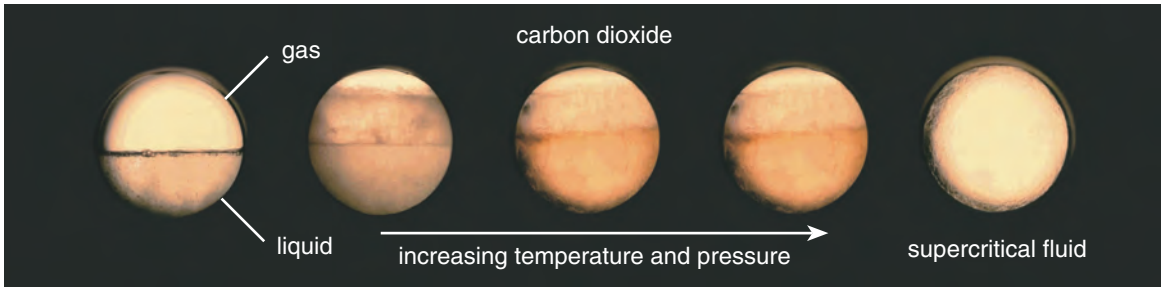


Fig. 52.10 Carbon dioxide changes to a supercritical fluid upon pressure and temperature change

A **supercritical fluid** is a substance above its critical point. A supercritical fluid shows the properties of both a liquid and a gas. At a pressure above 73.8 atmospheres and a temperature exceeding 31.1 °C (Fig. 52.9), carbon dioxide becomes a supercritical fluid.

Supercritical carbon dioxide — an environmentally benign solvent

Supercritical carbon dioxide is a good solvent for many non-polar organic compounds.

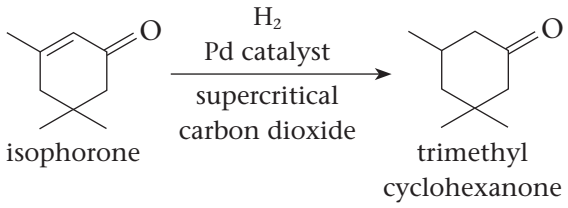
An advantage of supercritical carbon dioxide is that its viscosity is only about $\frac{1}{30}$ that of common liquid organic solvents. This enable reactants to migrate much faster through the fluid, thus speeding up their reactions. N8

N8 A supercritical fluid fills its container like a gas but with a density like that of a liquid.

Another main advantage of supercritical carbon dioxide is that this solvent can be easily turned into a gas by simply releasing the pressure. No separation process is required.

triple point 三相點　critical point 臨界點　supercritical fluid 超臨界流體

Hydrogenation is one type of well-studied synthetic processes in supercritical carbon dioxide. For example, supercritical carbon dioxide is used as a solvent in the hydrogenation of isophorone to produce trimethyl cyclohexanone.



isophorone — trimethyl cyclohexanone

52.7 Water — another environmentally benign solvent

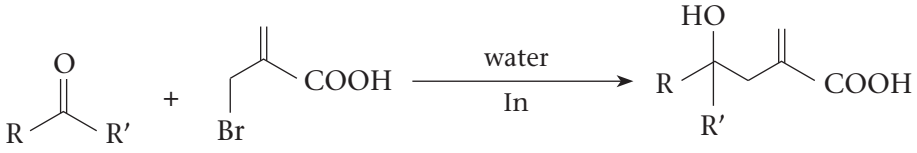
From a green chemistry viewpoint, the use of water as a solvent has many advantages:

- water is polar and thus relatively easy to separate from non-polar solvents;
- water is non-flammable;
- water is cheap and widely available;
- water is odourless and colourless — it is easy to recognize contamination;
- the density of water is different from that of most organic substances to allow convenient separation; and
- water has a high specific heat capacity, hence exothermic reactions may be controlled effectively.

Despite the advantages of using water as a solvent, the following reasons explain why it is not used that often:

- not all reactants dissolve in water;
- some reactants react with water and many catalysts are deactivated by reactions with water;
- water may be difficult to purify after reaction;
- the high specific heat capacity of water means that distillation requires high energy.

For example, addition reactions to carbonyl compounds frequently require a volatile organic compound as solvent and are relatively hazardous to carry out on an industrial scale. An alternative procedure using indium in water is now being developed.



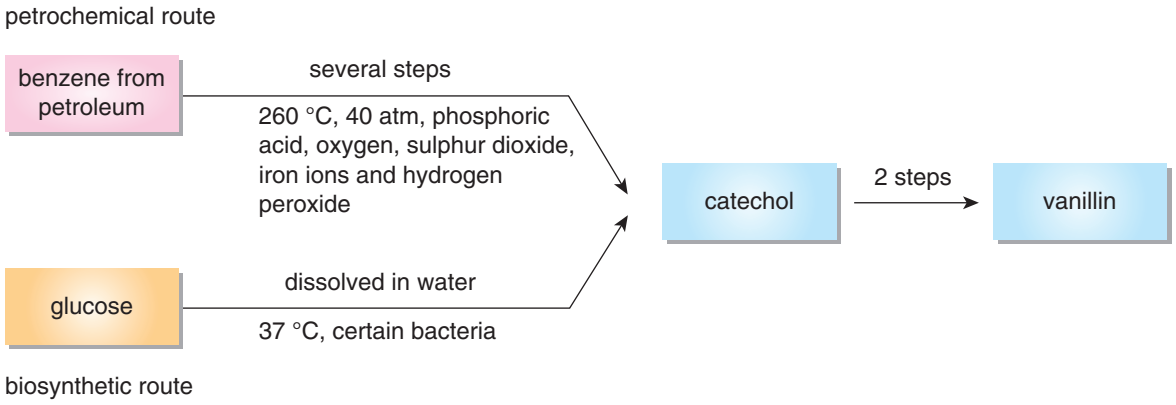
Worksheet
A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.



Discussion

Refer to page T64.

1 The following diagram shows two routes for the synthesis of the flavouring vanillin.



Explain why the biosynthetic route to vanillin is 'greener' than the petrochemical route.

2 The following three methods can be used to convert copper into copper(II) nitrate solution.

Method 1 $Cu(s) \longrightarrow CuO(s) \xrightarrow{\text{dilute } HNO_3(aq)} Cu(NO_3)_2(aq)$

Method 2 $Cu(s) \xrightarrow{\text{dilute } HNO_3(aq)} Cu(NO_3)_2(aq)$

Method 3 $Cu(s) \xrightarrow{\text{conc. } HNO_3(aq)} Cu(NO_3)_2(aq)$

a) Use chemical equations to describe the reactions in Method 1.

b) In Method 2, the reaction of copper with dilute nitric acid gives copper(II) nitrate, nitrogen monoxide and water. Write a chemical equation for this reaction.

CE 2006 Paper 1 Q7(b)

c) In Method 3, the reaction of copper with concentrated nitric acid gives copper(II) nitrate, nitrogen dioxide and water. Write a chemical equation for this reaction.

d) Work out the number of moles of HNO_3 required for producing 3 moles of $Cu(NO_3)_2$ via each method.

e) Suggest, with explanation, which method you would recommend for the conversion of copper into copper(II) nitrate solution.

CE 2006 Paper 1 Q7(c)

Worksheet

Two related worksheets with answers are available for download from *Jing Kung Chemistry Website*. Worksheet ① consists of additional questions for promoting active reading. (Note: Questions in this worksheet also cover those in the textbook.) Worksheet ②, requiring students to indicate the relationship among various terms by drawing, may help the less able readers improve their reading skills.

STSE Connections

The use of lead-free solder — making electronic materials greener

In recent years, industries and the general public have become increasingly aware of the importance of environmental issues and have paid full attention to green materials. Starting from July 2006, electronic equipment sold in the European Union has had to comply with the ROHS (reduction-of-hazardous-substances) directive. The ROHS directive seeks to prevent environmental damage by eliminating dangerous materials from electronic equipment that may eventually find their way into landfills and contaminate ground water. One of the restricted materials is lead, which manufacturers have widely used for more than 50 years as a prime component of solder and connection plating.



Fig. 52.11 Abandoned electronic equipment

Best low-temperature solder material — tin-lead alloy

Tin-lead alloy has been the most widely used solder material in electronic hardware for centuries. The tin-lead alloy has a melting point as low as 183 °C, an ideal solder material for welding electronic components. While research institutes around the world are searching for substitute alloys, the ones they have developed so far do not possess a melting point as low as that of tin-lead alloy. Take tin-silver alloy as an example; its melting point is 220 °C.

Global lead-free awareness

Industries are actively searching for possible formulae to produce lead-free products. Although every Printed Circuit Board (PCB) contains no more than 4% of lead, indiscreet disposal will lead to a considerable level of pollution.

In early 2006, China announced a law entitled 'Administration on the Control of Pollution Caused by Electronic Information Products', the so-called 'China ROHS'. China ROHS restricts specified substances (e.g. lead, mercury, cadmium, chromium(VI) ions) from electronic information products. China ROHS bears similarity to EU ROHS.

China ROHS applies to electronic information products for the Chinese market, irrespective whether manufactured in or imported into China.

Questions Refer to page T64.

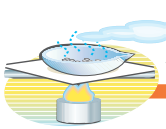
The elimination of lead is a global trend.

1 What challenges would scientists and engineers face?

2 What are the possible impacts on the manufacturers of domestic electronic equipment?

You may refer to the following websites for some ideas:

- An article discussing issues companies meet when trying to comply with the ROHS (reduction-of-hazardous-substances) directive
  https://www.gov.uk/rohs-compliance-and-guidance
- An article titled 'Lead-free and waste prevention requirements in the European Union: Cleaner products at lower cost'
  http://pdf.cloud.opensystemsmedia.com/advancedtca-systems.com/technology-in-europe-4.2005.pdf



Summary

1 Sustainable development is defined as 'development meeting the needs of the present without compromising the ability of future generations to meet their needs'.

2 Green chemistry is chemistry for the environment. It utilizes a set of principles that reduce or eliminate the use or generation of hazardous substances in the design, manufacture and use of chemical products.

3 Examples of applying green chemistry practices in the production of synthetic materials include:

- the use of reactants with little or no toxicity;
- the use of catalysts, such as phase-transfer catalysts;
- the use of safer solvents and auxiliaries, such as using supercritical carbon dioxide and water as solvents.

2 Any four of the following:
- Prevention
- Atom economy
- Less hazardous synthesis
- Designing safer chemicals
- Safer auxiliary substances
- Energy efficiency
- Use of renewable resources
- Reducing derivatives
- Catalysis
- Design for degradation
- Use real-time analysis for pollution prevention
- Accident prevention

Exercise

Note: The symbol 🌶 indicates the level of difficulty of a question.

Part I Knowledge and understanding

1 a) Explain the meaning of the term 'sustainable development'.

a) Sustainable development can be defined as 'development meeting the needs of the present without compromising the ability of future generations to meet their needs'.

b) What are the THREE components to sustainable development?
Economic growth, environmental protection and social equity.

2 Give FOUR principles of green chemistry.

3 Construct a concept map for the concepts learnt in this unit.
Refer to page T65.

4 a) Suggest THREE advantages of using water as a solvent in chemical processes.
Refer to page T66.

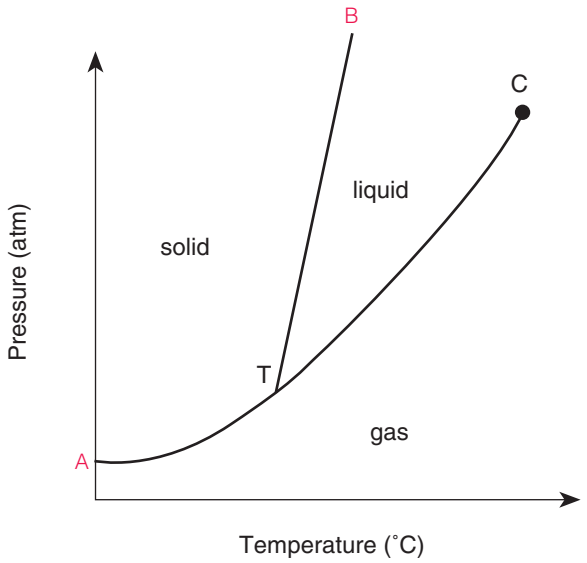
b) Suggest TWO reasons to explain why many chemical processes do not use water as a solvent.

5 Suggest THREE advantages of carrying out chemical processes without solvents.
Any three of the following:
- Faster reaction
- Lower energy usage
- Direct formation of high purity products
- Simplicity and low cost in equipment design

Part II Structured questions

Answers for the questions of the public examinations in Hong Kong are not provided.

6 The phase diagram for carbon dioxide is shown below.



a) Explain what the line TC shows.
The line TC shows the conditions of pressure and temperature under which the liquid and gas phases are in equilibrium.

b) What is the physical meaning of point C?
Point C is the critical point. Beyond this point, the liquid and gas phases become indistinguishable.

6 d) • The viscosity of supercritical carbon dioxide is much lower than that of many liquid organic solvents. Reactants migrate much faster through the fluid, thus speeding up their reactions.
• Supercritical carbon dioxide can be easily turned into a gas by releasing the pressure. Thus, no separation process is required.

c) Starting with a sample of $CO_2(g)$ at room temperature and pressure, suggest how solid CO_2 (dry ice) can be obtained. To obtain solid CO_2, increase the pressure and decrease the temperature so that the pressure is kept above line AT and the temperature kept below line TB.

d) Supercritical carbon dioxide can be used as a solvent in chemical processes. Suggest TWO advantages of using supercritical carbon dioxide.

7 Aluminium hydroxide is an active ingredient of antacid. Two paths for the production of aluminium hydroxide using Al(s), $H_2SO_4(aq)$ and NaOH(aq) as reactants are outlined below:

Path I: $Al(s) \longrightarrow Al_2(SO_4)_3(aq) \longrightarrow Al(OH)_3(s)$

Path II: $Al(s) \longrightarrow Na[Al(OH)_4](aq) \longrightarrow Al(OH)_3(s)$

a) Use chemical equations to describe the reactions in Path I and in Path II.

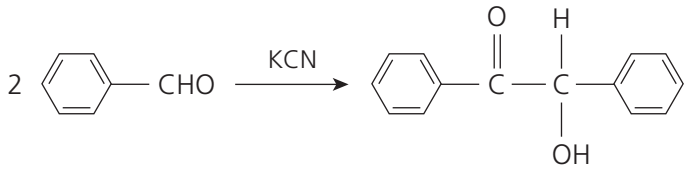
b) Work out the number of moles of H_2SO_4 and of NaOH required for producing 2 mol of $Al(OH)_3$ via Path I and via Path II.

c) Suggest, with explanation, whether Path I or Path II is recommended for the production of aluminium hydroxide.

(HKALE 2005)

Exercise

8 14.6 g of benzoin was obtained by heating 15.9 g of benzaldehyde with potassium cyanide in a solvent-free condition.



benzoin (m.p. 137 °C)

a) Calculate the percentage yield of benzoin from benzaldehyde.

b) Is this conversion an example of green chemistry? Explain.

c) Suggest an experimental procedure to isolate benzoin from the crude product.

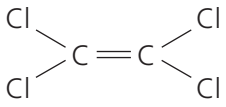
(Relative atomic masses: H = 1.0, C = 12.0, O = 16.0)

(HKASLE 2007)

9 Read the passage below and answer the questions that follow.

Greener dry cleaning

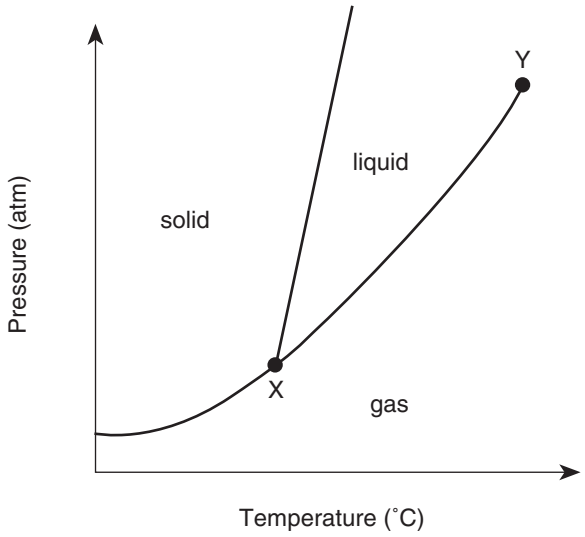
The term 'dry cleaning' is a little misleading since it is a liquid chemical solvent that removes the dirt and stains from clothing. The solvent used by most dry cleaners is tetrachloroethene.



Tetrachloroethene has serious environmental and health impacts. It is toxic to the liver and the central nervous system. It can accumulate in the body and is probably carcinogenic to humans. Tetrachloroethene is very persistent in ground water and soil.

One promising new approach to cleaning clothes is using supercritical carbon dioxide instead of tetrachloroethene. However, even though CO_2 is non-polar, it does not attract grease or oil strongly enough to remove them from clothing. Fortunately, it is found that a group of polymers can act as detergents in this solvent.

a) The phase diagram of carbon dioxide is shown below.



Identify the points X and Y. Point X is the triple point.
Point Y is the critical point.

b) Explain why a molecule of CO_2 has no net dipole moment although each C=O bond is polar.
A carbon dioxide molecule is linear in shape.
The individual C=O bond dipole moments cancel each other out exactly.
Thus, the molecule has no net dipole moment.

c) The structure of a typical anionic detergent can be represented as below:



where  represents a hydrocarbon tail

and  represents an anionic part attached to the hydrocarbon tail

The detergent can be used to clean a table cloth stained with grease. Explain the cleaning action of the detergent with reference to its structure.

d) From the perspective of green chemistry, describe TWO advantages of using carbon dioxide for dry cleaning.

Carbon dioxide is non-flammable and non-toxic.

e) Few dry cleaning companies have adopted carbon dioxide for dry cleaning at present. Suggest a possible reason.

Any one of the following:

- There is little awareness of the environmental and health impacts of tetrachloroethene used in dry cleaning.
- There is no legislative imperative to choose safer substitutes.

Exercise

9 c) When a detergent solution is added to a cloth with a greasy stain, the hydrocarbon 'tails' of the detergent particles dissolve in the grease. The anionic 'heads' of the detergent particles are insoluble in the grease and remain outside.
The surrounding water molecules attract the anionic 'heads' and lift the grease off the surface.
By stirring, the grease (which carries the dirt particles) breaks up into tiny droplets suspended in the water. These tiny droplets cannot come together again due to the repulsion between the anionic 'heads' of detergent particles on their surfaces. An emulsion forms.
Rinsing washes away the greasy suspension and leave the surface clean.